

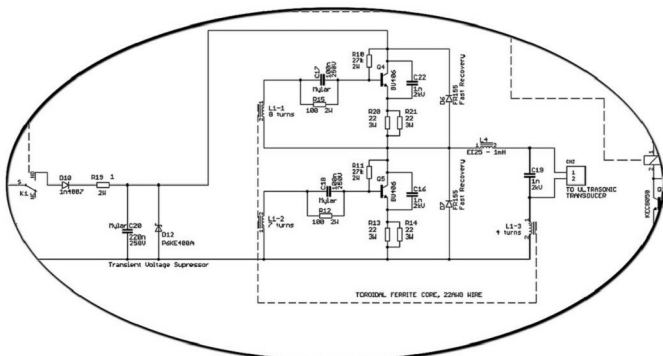


Fig. 6: Partial schematic of an ultrasonic cleaner with a clear view of transponder driver circuitry



Fig. 7: Removal of flux from printed circuit board using an ultrasonic cleaner

oscillator—an electronic relaxation oscillator that employs a saturable-core transformer. Fig. 6 shows the partial schematic of an ultrasonic cleaner with a clear view of the transponder driver circuitry.

Ultrasonic cleaning removes a variety of contaminants from parts immersed in aqueous media. Contaminants can be dirt, oil and grease, to name a few. Materials that can be cleaned include metals, glass, ceramics and so on. Ultrasonic cleaning is powerful enough to remove tough contaminants, yet gentle enough not to damage the substrate. It provides excellent pen-

etration and cleaning in the smallest crevices and between tightly-spaced parts in a cleaning tank.

Typical applications found in the metal industry are removing chips and cutting oils from cutting and machining operations,

removing buffing and polishing compounds prior to plating operations, and cleaning greases and sludge from rebuilt components for automotive and aircraft applications. Fig. 7 shows a typical application example—flux removal from printed circuit board using an ultrasonic cleaner.

## Give care

Efficiency of the core component, the transducer, affects both the cleaning time and efficacy achieved during the cleaning cycle. As the cleaning power of a unit stems from the transducer's performance, it is good to select a reputed-brand ultrasonic cleaner inte-

grated with reliable transducers.

Besides, there are other variables that need taking into account when cleaning items. Heat, power, frequency, detergent type and time influence the overall cleaning process. Also note that, the main electronic circuitry in most heated tabletop models is susceptible to damage from excess temperatures, so it should be set at optimal levels as recommended by the manufacturer.

Whenever possible, it is best to use a water based detergent in the ultrasonic cleaning process. Water is an excellent solvent. It is non-toxic, non-flammable and environmental-friendly. Cavitation energy in a water based solution is more intense than in an organic solvent.

Similarly, temperature of the solution has a profound effect on ultrasonic cleaning potential. Higher temperatures will result in higher cavitation intensity and better cleaning. However, if the temperature approaches the boiling point of the solution (water cavitates most effectively at about 70°C), the liquid will boil in the negative pressure areas of the sound waves, reducing or eliminating cavitation. **EFY**